INT 422
Pattern Recognition
2024

Dr. Maysa Khalil Noby
Lecturer of Information Technology & Computer Science
Maysa.khalil@su.edu.eg

sinaiuniversity.net

Lecture 10: K-Mean Clustering Algorithm

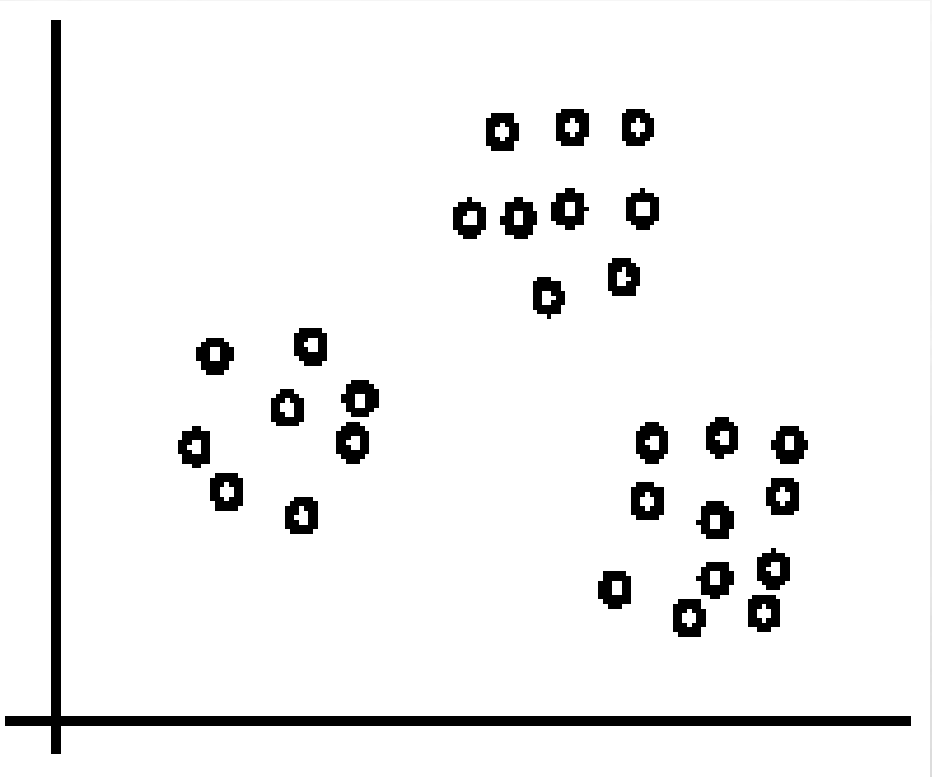
Agenda

- What is Clustering?
- What is a Good Clustering?
- Clustering Algorithms
- Hard vs. soft clustering
- Clustering Approaches
- K-Means Algorithm
- Termination conditions
- K-Means Example

What is Clustering?

- Cluster: a collection of data objects
  - Similar to one another within the same cluster
  - Dissimilar to the objects in other clusters
- Cluster analysis
  - Grouping a set of data objects into clusters
- Clustering is unsupervised classification: no predefined classes

What is Clustering?



What Is a Good Clustering?

- A good clustering method will produce clusters with
  - High intra-class similarity
  - Low inter-class similarity

Precise definition of clustering quality is difficult

- Application-dependent
- Ultimately subjective

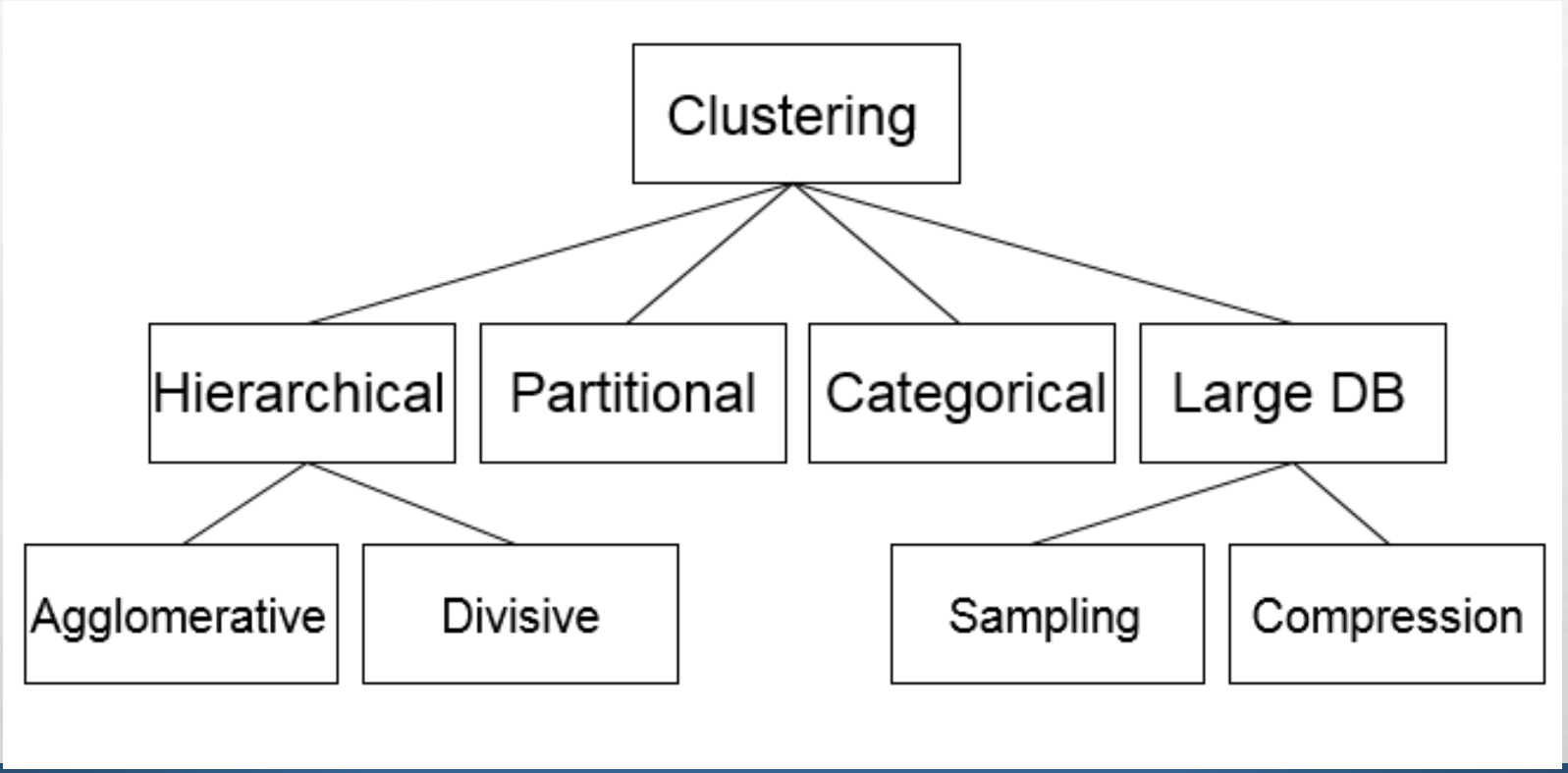
Clustering Algorithms

- **Flat algorithms**
  - Usually start with a random (partial) partitioning
  - Refine it iteratively
    - *K* means clustering
    - (Model based clustering)
- **Hierarchical algorithms**
  - Bottom-up, agglomerative
  - (Top-down, divisive)

Hard vs. soft clustering

- **Hard clustering:** Each object belongs to exactly one cluster
  - More common and easier to do
- **Soft clustering:** An object can belongs to more than one cluster.
  - Makes more sense for applications like creating browsable hierarchies
  - You may want to put a pair of sneakers in two clusters: (i) sports apparel and (ii) shoes
  - You can only do that with a soft clustering approach.

Clustering Approaches



K-Means Algorithm

- K means clustering is a popular machine learning algorithm. It's an unsupervised method because it starts without labels and then forms and labels groups itself. K means clustering is not a supervised learning method because it does not attempt to predict existing or known group labels.

K means" refers to the following:

- The number of clusters you specify (K).
- The process of assigning observations to the cluster with the nearest center (mean).
- K means clustering forms the groups in a manner that minimizes the variances between the data points and the cluster's centroid.

K-Means Algorithm

Select K random objects $\{s_1, s_2, \ldots s_K\}$ as seeds.

Until clustering converges (or other stopping criterion):

For each doc d_i:

Assign d_i to the cluster c_j such that $dist(x_i, s_j)$ is minimal.

(Next, update the seeds to the centroid of each cluster)

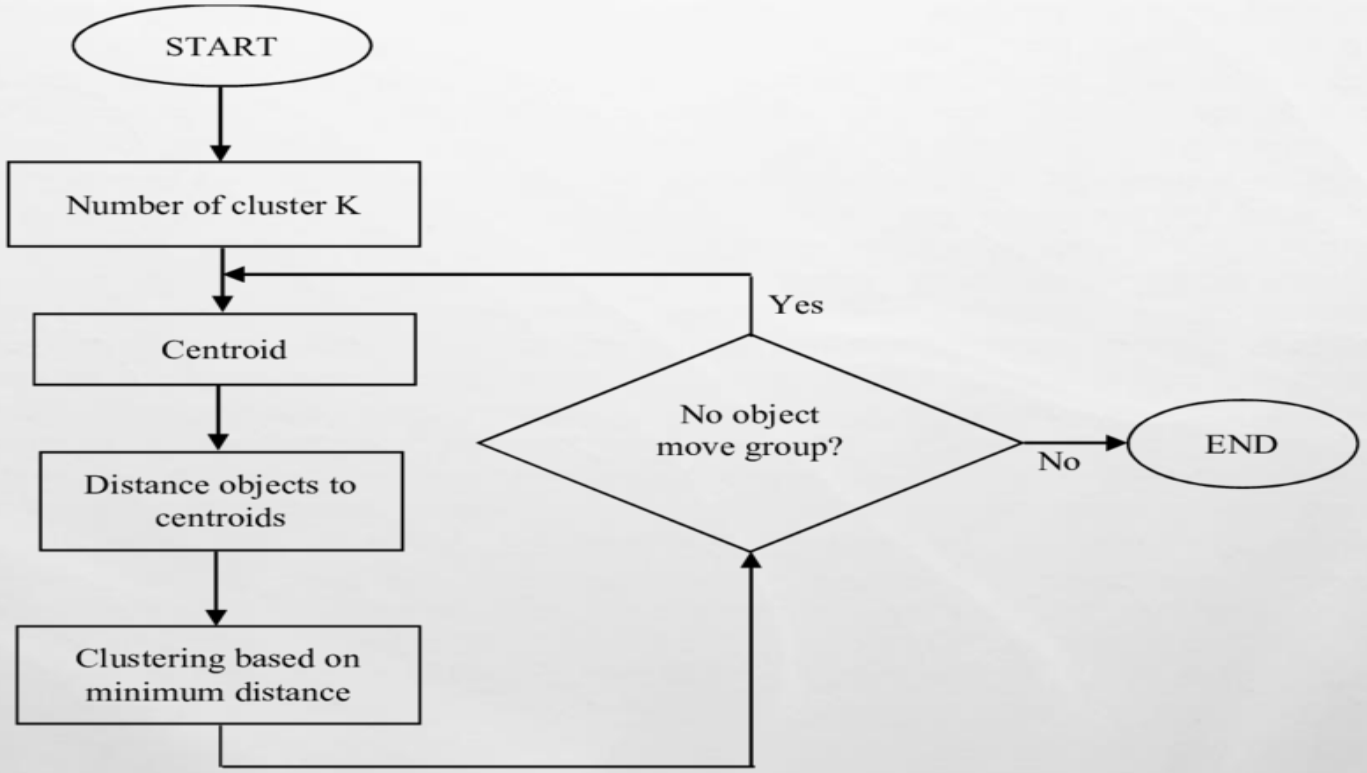
For each cluster c_j

$s_j = \mu(c_j)$

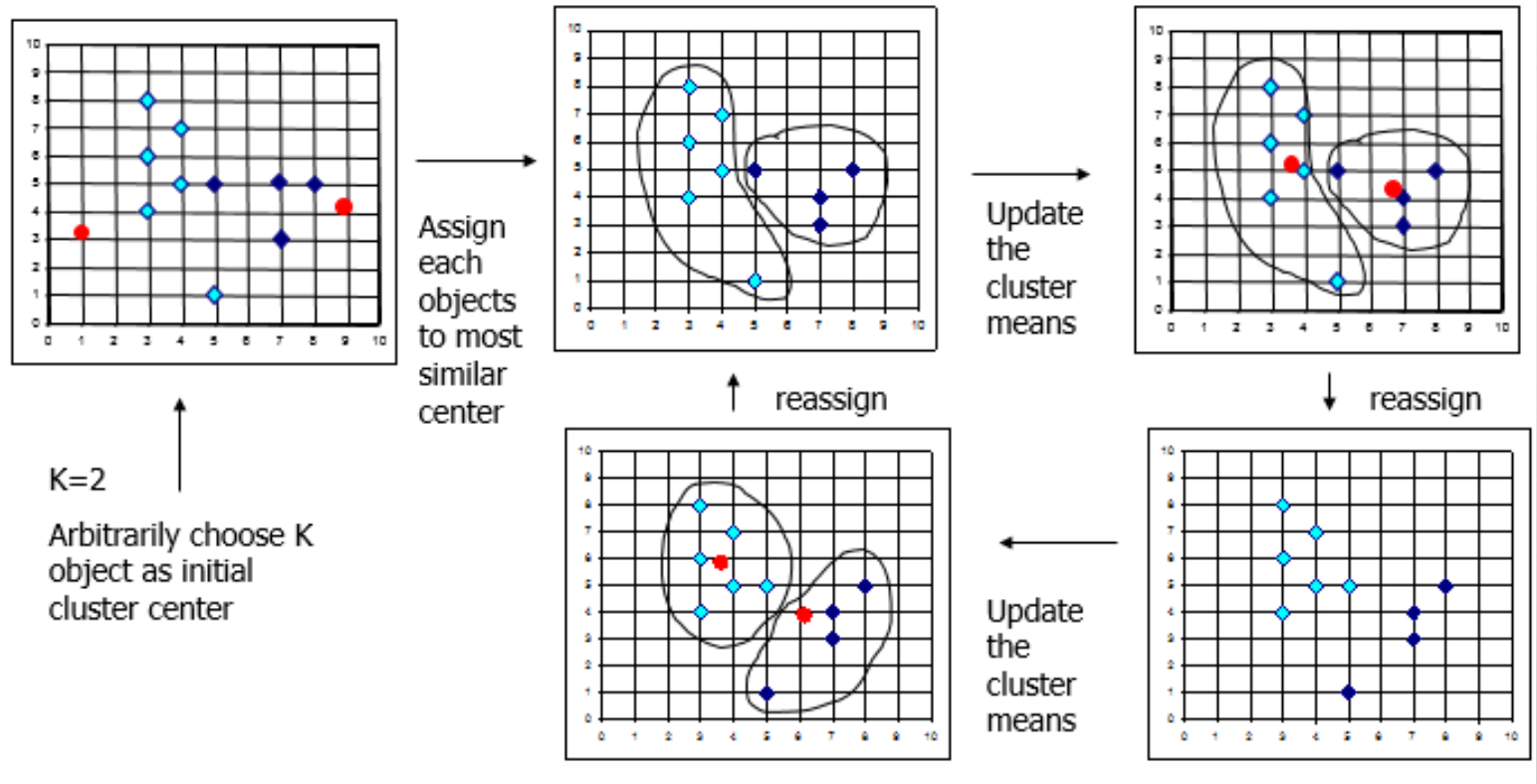
Questions related to K-Mean

- What is k-means trying to optimize?
- Will k-means stop (converge)?
- Will it find a global or local optimum?
- How to pick starting cluster centers?
- How many clusters should we use?

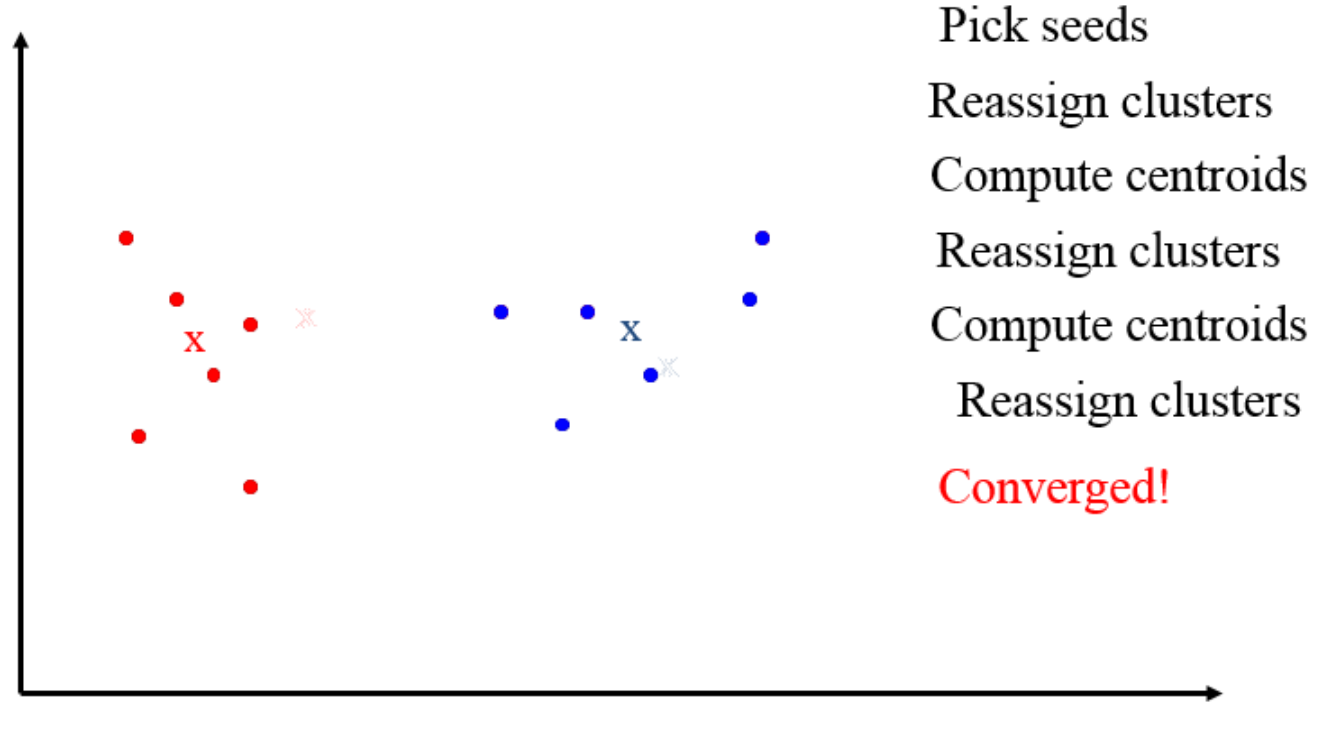
K-Means Algorithm



K-Means Algorithm



K Means Example (K=2)



Termination conditions

- **Several possibilities, e.g.,**
  - A fixed number of iterations.
  - Object partition unchanged.
  - Centroid positions don’t change.

K-Mean Example 1

Cluster the following eight points (with (x, y) representing locations) into three clusters A1(2, 10) A2(2, 5) A3(8, 4) A4(5, 8) A5(7, 5) A6(6, 4) A7(1, 2) A8(4, 9).

Initial cluster centers are: A1(2, 10), A4(5, 8) and A7(1, 2).

The distance function between two points $a=(x1, y1)$ and $b=(x2, y2)$ is defined as: $\rho(a, b) = |x2 - x1| + |y2 - y1|$.

- Use k-means algorithm to find the three cluster centers after the second iteration.

$$\sqrt{(X_2 - X_1)^2 + (Y_2 - Y_1)^2}$$

K-Mean Example 1

Iteration 1

		(2, 10)	(5, 8)	(1, 2)	
	Point	**Dist Mean 1**	**Dist Mean 2**	**Dist Mean 3**	**Cluster**
A1	(2, 10)				
A2	(2, 5)				
A3	(8, 4)				
A4	(5, 8)				
A5	(7, 5)				
A6	(6, 4)				
A7	(1, 2)				
A8	(4, 9)				

K-Mean Example 1

Iteration 1

		(2, 10)	(5, 8)	(1, 2)	
	Point	**Dist Mean 1**	**Dist Mean 2**	**Dist Mean 3**	**Cluster**
A1	(2, 10)	0	5	9	1
A2	(2, 5)	5	6	4	3
A3	(8, 4)	12	7	9	2
A4	(5, 8)	5	0	10	2
A5	(7, 5)	10	5	9	2
A6	(6, 4)	10	5	7	2
A7	(1, 2)	9	10	0	3
A8	(4, 9)	3	2	10	2

Cluster 1
(2, 10)

Cluster 2
(8, 4)
(5, 8)
(7, 5)
(6, 4)
(4, 9)

Cluster 3
(2, 5)
(1, 2)

K-Mean Example 1

- Next, we need to re-compute the new cluster centers (means). We do so, by taking the mean of all points in each cluster.
- For Cluster 1, we only have one point A1(2, 10), which was the old mean, so the cluster center remains the same.
- For Cluster 2, we have ((8+5+7+6+4)/5, (4+8+5+4+9)/5) = (6, 6)
- For Cluster 3, we have ((2+1)/2, (5+2)/2) = (1.5, 3.5)

K-Mean Example 1

Iteration 2

		(2, 10)	(6, 6)	(1.5, 3.5)	
	Point	Dist Mean 1	Dist Mean 2	Dist Mean 3	**Cluster**
A1	(2, 10)	0	8	7	1
A2	(2, 5)	5	5	2	3
A3	(8, 4)	12	4	7	2
A4	(5, 8)	5	3	8	2
A5	(7, 5)	10	2	7	2
A6	(6, 4)	10	2	5	2
A7	(1, 2)	9	9	2	3
A8	(4, 9)	3	5	8	1

K-Mean Example 1

- Next, we need to re-compute the new cluster centers (means). We do so, by taking the mean of all points in each cluster.

- In Cluster 1, we have points 1 and 8. Therefore the centroid is: ((2+4)/2,(10+9)/2)=(3,9.5)
- In Cluster 2, we have points 3,4,5 and 6. Therefore, the centroid is: ((8+5+7+6)/4,(4+8+5+4)/4)=(6.5,5.25)
- For Cluster 3, we have points 2 and 7. Therefore, the centroid is: ((2+1)/2,(5+2)/2)=(1.5,3.5)

K-Mean Example 1

Iteration 3

		(3, 9.5)	(6.5 ,5.25)	(1.5, 3.5)	
	Point	Dist Mean 1	Dist Mean 2	Dist Mean 3	**Cluster**
A1	(2, 10)	1.5	9.25	7	**1**
A2	(2, 5)	5.5	4.75	2	**3**
A3	(8, 4)	10.5	2.75	7	**2**
A4	(5, 8)	3.5	4.25	8	**1**
A5	(7, 5)	8.5	0.75	7	**2**
A6	(6, 4)	8.5	1.75	5	**2**
A7	(1, 2)	9.5	8.75	2	**3**
A8	(4, 9)	1.5	6.25	8	**1**

K-Mean Example 1

- Next, we need to re-compute the new cluster centers (means). We do so, by taking the mean of all points in each cluster.

- In Cluster 1, we have points 1, 4, and 8. Therefore the centroid is: $((2+5+4)/2,(10+8+9)/2)=(3.67,9)$
- In Cluster 2, we have points 3,5 and 6. Therefore, the centroid is: $((8+7+6)/4,(4+5+4)/4)=(7,4.3)$
- For Cluster 3, we have points 2 and 7. Therefore, the centroid is: $((2+1)/2,(5+2)/2)=(1.5,3.5)$

K-Mean Example 1

Iteration 4

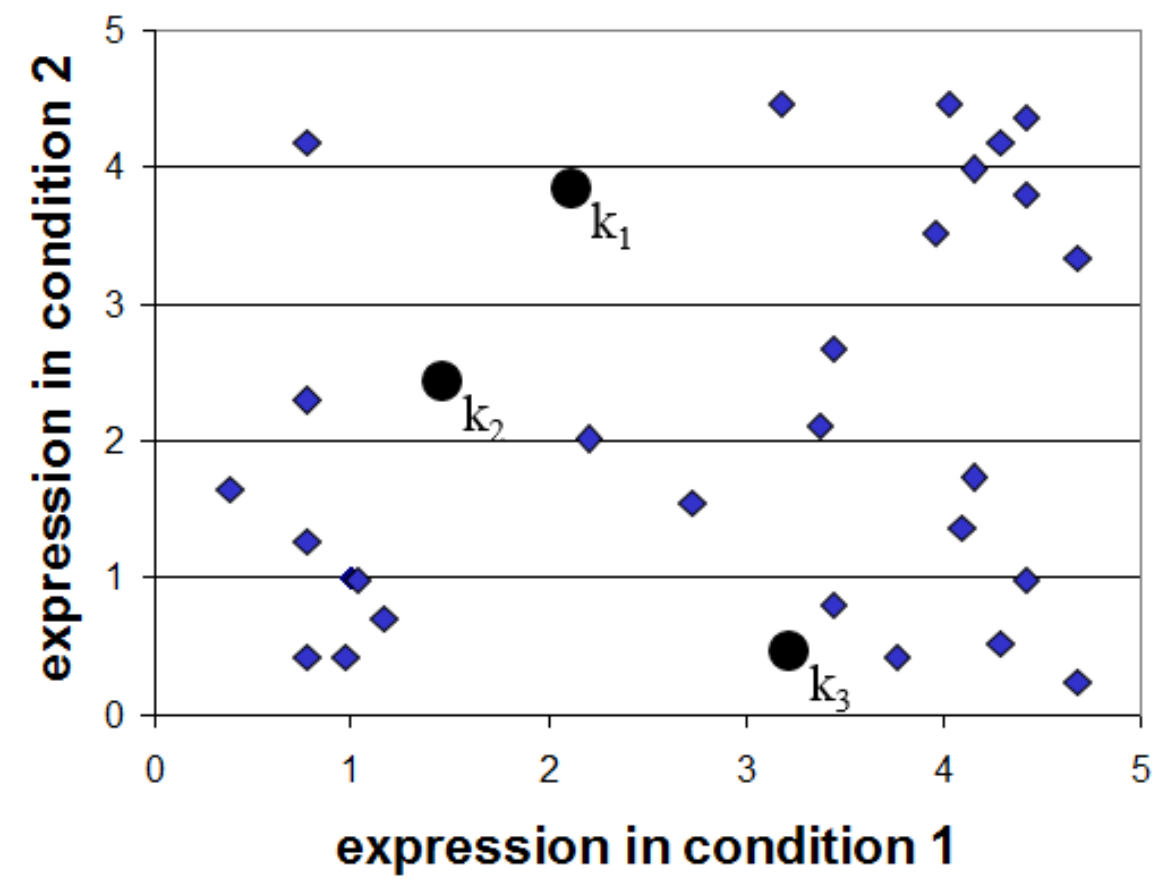
		(3.67, 9)	(7 ,4.3)	(1.5, 3.5)	
	Point	Dist Mean 1	Dist Mean 2	Dist Mean 3	**Cluster**
A1	(2, 10)	2.67	10.7	7	**1**
A2	(2, 5)	5.67	5.7	2	**3**
A3	(8, 4)	9.33	1.3	7	**2**
A4	(5, 8)	2.33	5.7	8	**1**
A5	(7, 5)	7.33	0.7	7	**2**
A6	(6, 4)	7.33	1.3	5	**2**
A7	(1, 2)	9.67	8.3	2	**3**
A8	(4, 9)	0.33	7.7	8	**1**

K-Mean Example 1

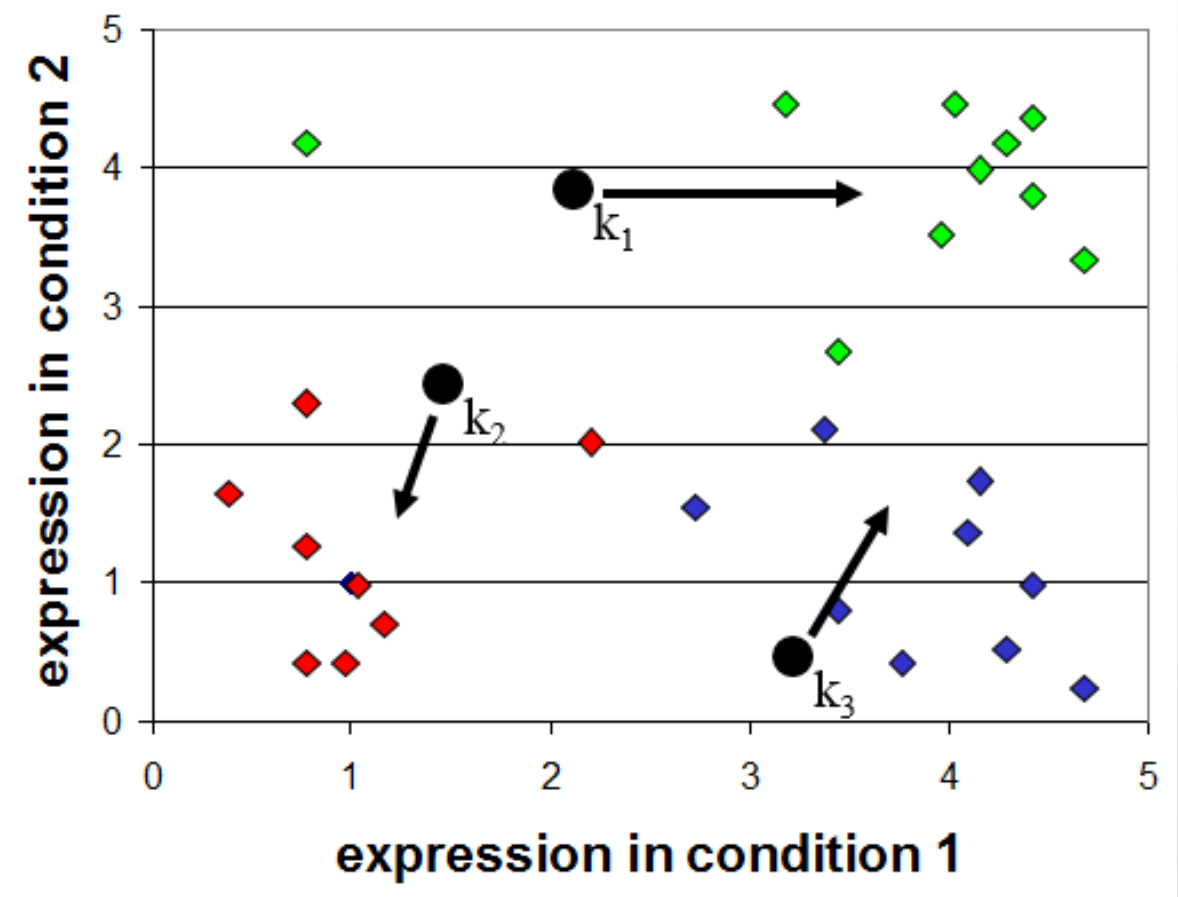
- The termination is occurred as the results of iteration 3 and 4 is the same.
- The results are:
  - Cluster 1 contains A1, A4, A8
  - Cluster 2 contains A3, A5, A6
  - Cluster 3 contains A2, A7

K Means Example (K=3)
termination condition is 3 iterations

iteration # 1

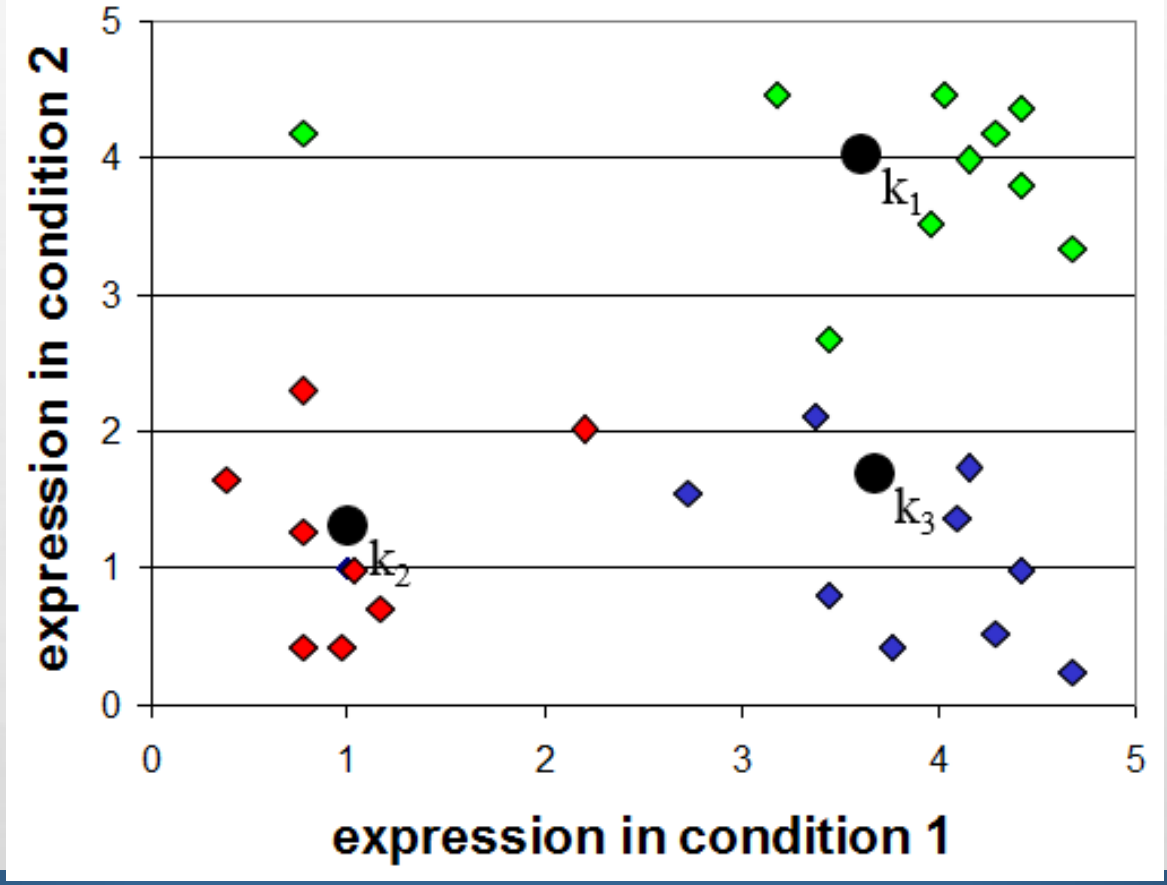


K Means Example (K=3)
termination condition is 3 iterations

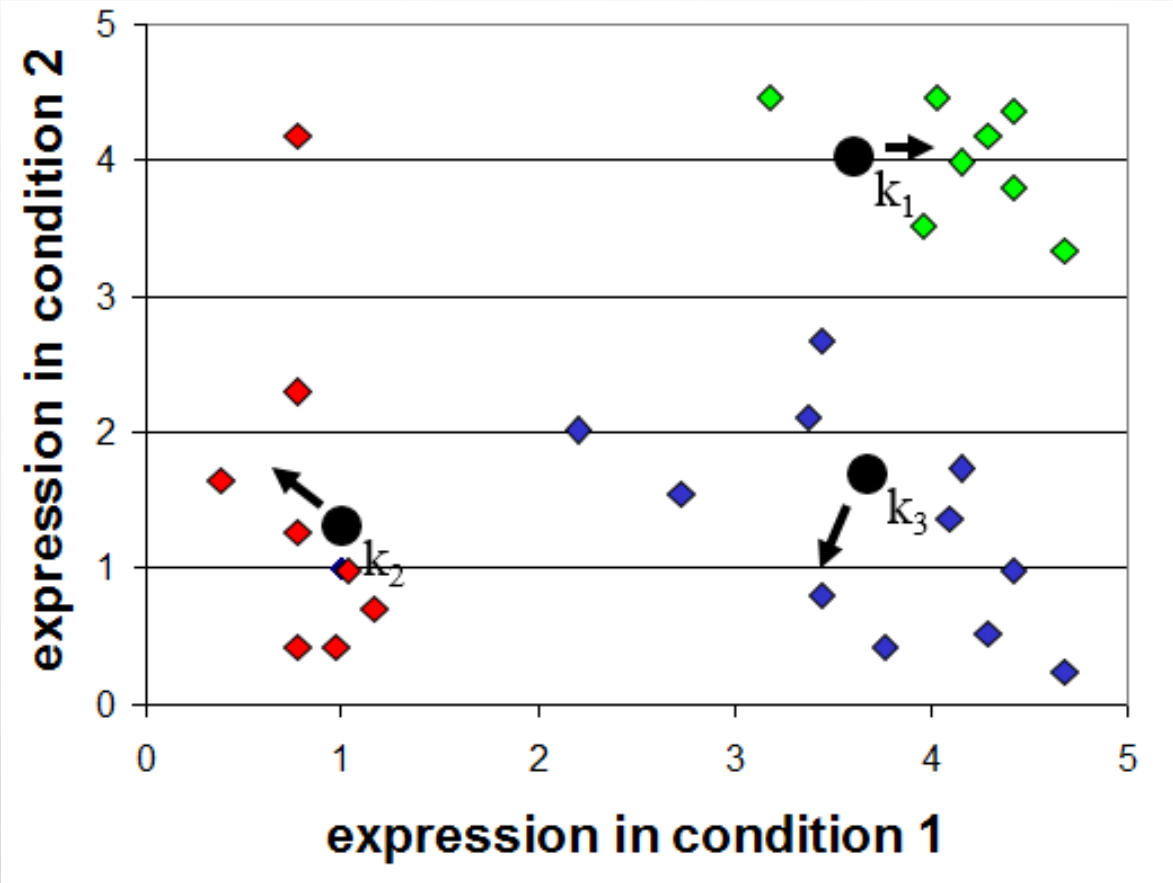


K Means Example (K=3)
termination condition is 3 iterations

iteration # 2

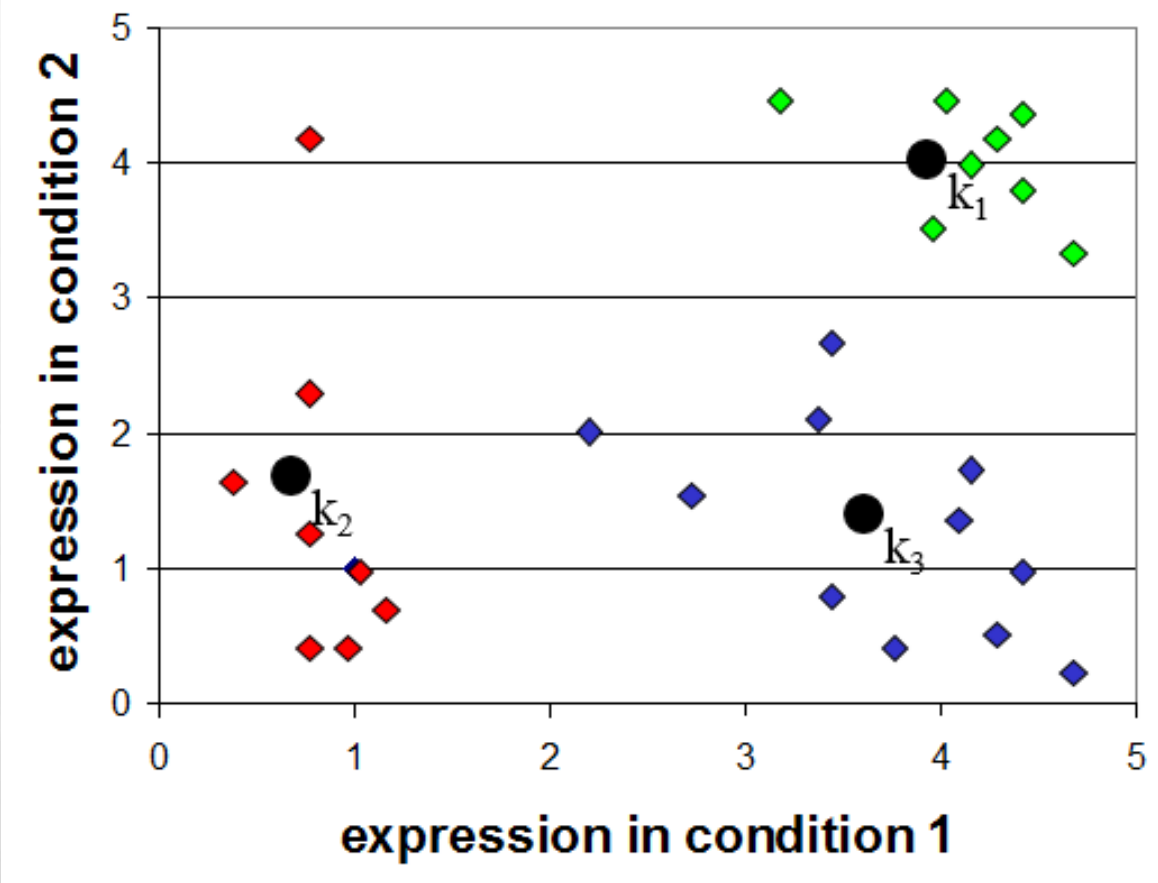


K Means Example (K=3)
termination condition is 3 iterations



K Means Example (K=3)
termination condition is 3 iterations

iteration # 3



Pros & Cons of K-Mean Algorithm

Advantages of k-means

1. Relatively simple to implement.
2. Scales to large data sets.
3. Guarantees convergence.
4. Easily adapts to new examples.

Disadvantages of k-means

1. Choosing \(k\) manually.
2. Being dependent on initial values.
3. Scaling with number of dimensions.

Review Questions

- What is a good clustering?
- When K-mean clustering is terminated?
- What is the difference between Hard vs. soft clustering?
- Explain the K-mean algorithm.
- Mention the Cons and pros of the K-mean algorithm.

Assignment 1

	X	Y
1	25	79
2	27	78
3	33	74
4	34	57
5	54	32
6	59	35
7	52	58
8	47	50
9	35	14
10	43	27

The initial points are

- k1(30,40)
- k2(40,55)

Use (K=2) for 3 iterations

Assignment 2

No	X	y	Class
1	185	72	?
2	170	56	?
3	168	60	?
4	179	68	?
5	182	72	?
6	188	77	?

The initial points are

K1	185	70
K2	175	65

Use (K=2) for 3 iterations

Assignment 3

Individual	Variable 1	Variable 2
1	1	1
2	1.5	2
3	3	4
4	5	7
5	3.5	5
6	4.5	5
7	3.5	4.5

Assume the first 2 centroid are 1 and 4

Questions ?

Thank You